

Algorithmic thinking: iteration

Revise 360

Worksheet for the 360° experience: Loop planning lab

Name: _____

Class: _____

Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Loop planning lab. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Choose count- or condition-controlled iteration.
- Trace loops and running totals.
- Explain why loops terminate.

Starter: before you put the headset on

A task repeats exactly five times. Would a counter or an unknown stopping condition be clearer?

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Key terminology

Iteration

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Accumulator

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The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 Repeat a known count

Turn right

Key facts in your own words:

Challenge: How many times does this loop print?

OCR pseudocode: for i = 1 to 4. How many iterations?

OCR for i = 2 to 5 executes...

2 Repeat until a condition

Turn right

Key facts in your own words:

Challenge: When does the loop stop?

A while condition is false before the first check. How often does its body run?

Which loop can execute zero times?

3 Accumulate a total

Turn around

Key facts in your own words:

Challenge: What goes wrong if `total = 0` is inside the loop?

Starting at 0, add 5, 2 and 8. What is the total?

Where should a running total normally be initialised?

4 Guarantee progress

Turn around

Key facts in your own words:

Challenge: What happens if the update is removed?

What fixes a loop whose counter never changes?

`x = 0, while x < 3, x = x + 1. Final x?`

5 Selection inside a loop

Turn left

Key facts in your own words:

Challenge: How many of 2, 7, 8 and 3 are at least 7?

Which values pass value ≥ 5 in [1, 5, 8]?

Count values > 5 in [5, 6, 2, 8]. Count?

6 Nested loops

Turn left

Key facts in your own words:

Challenge: How many prints for outer 4 and inner 3?

Outer loop runs twice. Inner loop runs five times each time. Total inner executions?

Outer 3, inner 4, both fixed counts. Inner runs?

★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

Order a running-total algorithm.

Put these steps in order: Output total after the loop / Add the value to total inside the loop / Repeat for each input value / Set total to 0

A loop starts $n = 1$, adds 1 while $n < 4$. What is n after the loop?

Your score

Copy it from the score screen as soon as it appears.

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Key question

How could a program process 100 loans without writing the same instructions 100 times?

Review and improve

Use review mode. Explain one correction and why the correct answer fits. In Python, write and run a program that inputs three numbers and prints their total. Test 2, 3, 4 (expected 9) and -2, 2, 0 (expected 0).

Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) Explain one difference between count-controlled and condition-controlled loops. [2]

b) Write pseudocode to input three numbers and output their total. [4]

c) Why might a while loop never terminate? [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Choose count- or condition-controlled iteration.			
Trace loops and running totals.			
Explain why loops terminate.			